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ReaxFF molecular dynamics simulation and experimental validation about chemical reactions of water and alcohols on SiC surface

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ABSTRACT

The high chemical inertness of silicon carbide (SiC) makes it difficult for surface chemical reactions to occur. Surface oxidation reactions are crucial for improving the processing efficiency and surface quality of SiC wafers. Extensive research has been conducted on conventional aqueous polishing slurries, while investigations on non-aqueous solvent-based polishing slurries have emerged as alternative options for enhancing SiC surface processing, as evidenced by recent scholarly reports. However, the surface chemical reaction mechanism of silicon carbide in either water or other hydroxyl groups containing solvents remains still unclear and is not well understood. In this work, ReaxFF molecular dynamics simulation was used to study the dynamic reaction processes between solvents and 6H SiC (001) surfaces. The oxidation of SiC surface in water and alcohol solvent is found to go through three stages: in the first stage, the solvent molecules move towards the SiC surface and react with the uncoordinated Si atoms and C atoms on the surface; in the second stage, the solvent molecules adsorbed on the surface of SiC reduces the *Si*–C bonding energy and promotes the breaking of *Si*–C bonds; in the third final stage, the migration of OH and H on the SiC surface induces the formation of *Si*–*O*–Si bonds, which promotes the continuous oxidation of the SiC surface. The order of simulated chemical reactivity between solvent and silicon carbide surface is consistent with experimental polishing removal rates. The reaction mechanism of solvent on silicon carbide surface was proposed by combining ReaxFF simulation results with polishing experiments, AFM, and XPS data analysis, shedding new insights on better understanding the mechanism of SiC chemical mechanical polishing.

1. Introduction

Silicon carbide (SiC) devices have found extensive application in the electric automotive, windmill transformer, aerospace, nuclear energy, and semiconductor sectors, owing to their superior thermo-mechanical characteristics, remarkable chemical corrosion resistance, and extraordinary performance as wide band-gap semiconducting materials [1–3]. The conductive 6H–SiC compound possesses considerable potential for application in high-temperature environments as a radiation-resistant and high-power optoelectronic material [4]. The Si and C atoms in SiC exhibit a robust interatomic covalent bonding, yielding remarkably high hardness, chemical inertness, and superior thermal conductivity [5]. These properties are advantageous for various applications; however, they pose challenges in the processing of SiC [6,7]. In chemical mechanical polishing (CMP), the removal of SiC surface materials is considered to be dominated by friction chemical reactions [8]. The inherent surface chemical characteristics of materials were identified as paramount for the tribological performance [9–13], a pivotal subject matter in this investigation.

The surface species adsorbed on the surface of the material affect

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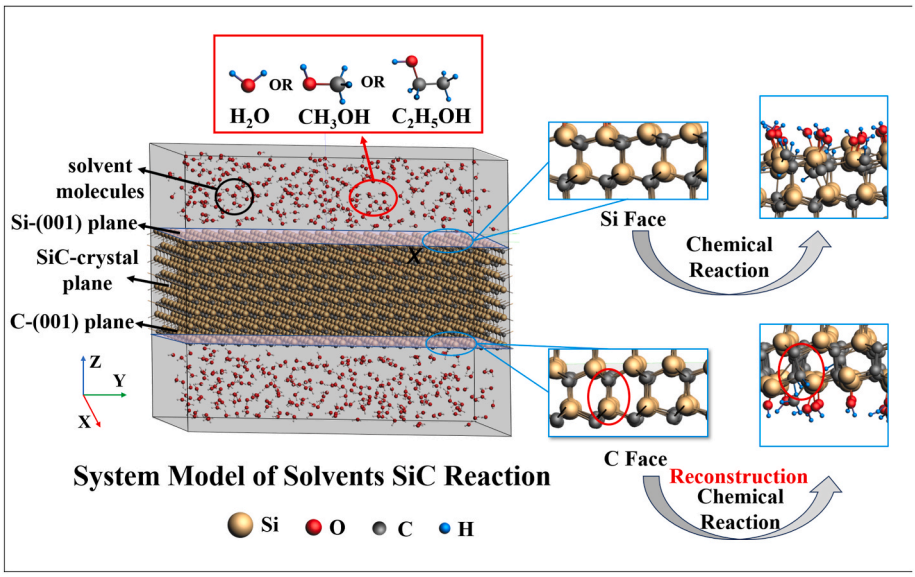

Fig. 1. Side view of SiC and solvent system, which contains Si, O, C, H.

their hydrophilicity/hydrophobicity and the formation of *Si*–*O*–Si bonds on the surface, thereby affecting the friction and wear of the material in the tribological reaction [14,15]. Much research has been done on the effect of surface chemical species (i.e., *Si*–H, Si–OH, and *Si*–*O*–Si) on silicon-based materials on their tribological behavior in aqueous environments [9,14–24]. Based on these investigations, it is widely acknowledged that water molecules can be adsorbed onto silicon surfaces. Some of these water molecules undergo dissociation, resulting in the formation of H atoms and OH groups, a phenomenon known as dissociative adsorption. Conversely, certain adsorbed water molecules remain intact without dissociation, a process referred to as molecular adsorption. The surface of the silicon-containing material was terminated with H_2O, H, and OH species. The silicon surface terminates with H species, as *Si*–H is hydrophobic, and is hydrophilic [25] when it is covered with Si–OH groups. Hydrophobic silicon surfaces exhibit high friction and wear [9], which is beneficial for the mechanical removal of material. The procedure of dehydration and dehydrogenation reactions occurring between the Si–OH groups on the surface and the adjoining Si atoms can lead to the formation of cohesive *Si*–*O*–Si bonds at the interface [20,21]. These previous studies suggest that the formation of interfacial *Si*–*O*–Si bonds plays a key role in the surface wear of Si. Nevertheless, various other simulations conducted on the subject of friction have revealed that the migration of H atoms across the contact interface and into the subsurface region [26,27], as well as the incorporation of O atoms into the friction surface through the formation of OH groups [28], can also result in bond rupture and degradation of the material.

Unlike silicon-based materials, SiC is much more chemically inert, and chemical reactions are more difficult to take place on SiC surfaces [5,8,29,30]. Many studies have been conducted on the friction and wear of the SiC surface [4,29–38], and the formation of interfacial *Si*–*O*–Si bonds was also found to play an important role in the surface wear of SiC materials. However, these studies mainly focus on the mechanical wear of SiC and the thermal oxidation mechanism of SiC materials. So far, few studies have been made on the effect of chemical functional groups on the surface of SiC materials on the oxidation of SiC surfaces, especially in non-aqueous solution systems.

Methanol and ethanol solvents have been revealed in the previous report to have improved material removal rate (MRR) and significantly better surface quality than aqueous solvents in a fixed abrasive polishing of SiC [39]. In the present investigation, molecular dynamics simulations utilizing the ReaxFF method have been employed to examine the dynamic reaction mechanism between the solvent and the 6H–SiC (001) surface at various temperatures. The surface of SiC (001) was exposed to three solvents, namely water, methanol, and ethanol, at temperatures of 300 K, 1000 K, and 2000 K to investigate the interfacial physical and chemical actions of solvent molecules on the exposed SiC (001) surface at varied temperatures. Due to its adeptness in modeling bond formation and dissociation, as well as simulating reactive systems with a significant atom count [40], the ReaxFF reactive force field proves to be a fitting method for examining the adsorption and dissociation mechanisms of solvent molecules on 6H–SiC surfaces. This assertion is supported by robust research in the field [34]. To validate the simulation results, polishing experiments, AFM measurement of surface roughness, and XPS surface composition analysis after solvent corrosion of SiC were conducted. The interaction mechanism between water and alcohol solvents on the surface of SiC was proposed based on experimental and simulation results to further our understanding of the chemical nature of polishing slurries on the SiC surface and to provide guidance on developing an effective and more economically viable polishing system for SiC wafer manufacturing.

2. Simulation and experimental methods

2.1. ReaxFF reactive force field method

This study employs molecular dynamics simulations utilizing the ReaxFF reaction force field to examine the reaction mechanism between the SiC crystal surface and the solvent. For the purpose of this investigation, the ReaxFF reactive force field and three-dimensional periodic boundary conditions were employed in X, Y, and Z directions to conduct molecular dynamics simulations in order to explore the reaction mechanisms occurring between the SiC crystal surface and solvent. Through the utilization of this multiscale simulation method, which effectively combines the MD method and the finite element method, an atomic description of the interface and a continuous description in the bulk was achieved, thereby enhancing the accessible length scales [41]. The potential function served as a key tool to describe molecular interactions, with the interatomic forces being derivable from its derivatives. Notably, the molecular dynamics (MD) modeling approach was highly suitable for exploring various aspects such as nanoscale cutting, interface interactions, crystal orientations, and cutting directions [42]. This was achieved to bridge the gap that exists between MD simulations and the physical micro/nano polishing processes while

Table 1
Initial configurations of solvent and SiC surfaces.

3D coordinate system	H_2O	CH_3OH	C_2H_5OH
X (Å)	30.8	30.8	30.8
Y (Å)	61.6	61.6	61.6
Z (Å)	30.0	35.8	44.5
t (Å)	18.3	18.3	18.3
α (°)	90	90	90
β (°)	90	90	90
γ (°)	120	120	120
N1	597	600	598
N (C)	200	200	200
N (Si)	200	200	200

taking into consideration the macro scale of the wafer and the micro/-nanoscale of the wafer surface (3D topography). In contrast to nonreactive force fields, ReaxFF employs a universal correlation between bond order and bond distance, as well as between bond energy and bond order, enabling an accurate representation of bond formation and dissociation during the molecular dynamics simulations. The ReaxFF reaction molecular dynamics simulation technique not only demonstrates its ability to elucidate the intricate interplay between water and metal/metal oxide its portrayed outcomes in academic literature [24,43–45], but also presents satisfactory results in simulating the interaction linking water, crystalline silicon, and amorphous silica [9,16–19]. Moreover, this methodology can be effortlessly expanded to encompass any framework comprised of diverse substances. We adopted the ReaxFF force field [34] developed based on the Si/C/O/H/N/S force combination to deal with the relationship between Si, O, C, and H atoms to properly describe SiC and the relationship between water and alcohol.

2.2. Computational setup

To reveal the reaction process between the 6H–SiC surface and solvents such as H_2O, CH_3OH, and C_2H_5OH, we established a model system comprising a SiC (001) substrate and a solution of H_2O (or CH_3OH, C_2H_5OH) to simulate the aforementioned interaction. In order to achieve a sensible density, we undergo a relaxation process within the NPT ensemble. Within a relaxation period, time of approximately 50 Ps, the system's volume can attain a state of relative stability. The model build-up and calculation process is conducted as follows: the 6H–SiC unit cell consisting of 1600 Si atoms and 1600 C atoms is established through the software database, and the 001 crystal plane is exposed to form the SiC substrate orientation. Solvent molecules were added to the model cells, using the NPT ensemble with a pressure set at 1 atm. After undergoing equilibration at a temperature of 50 K for a period of 50 Ps, with the aid of a Berendsen [46] thermal bath possessing a damping constant of 10 fs, the system is brought to a minimum potential energy, thereby forming the ultimate model, illustrated in Fig. 1. The intricacies of these models are clearly outlined in Table 1.

(t is the thickness of the SiC substrate, α represents the angle between the X-axis and the Z-axis in the unit coordinate system, while β denotes the angle between the Y-axis and the Z-axis in the same coordinate system. γ, on the other hand, signifies the angle between the X-axis and the Y-axis. N1 represents the overall quantity of solvent molecules in system. N (C) denotes the quantity of 1 monolayer (ML) carbon atoms, while N (Si) stands for the quantity of 1 monolayer (ML) silicon atoms.)

Table 1 illustrates the prearranged dimensions (X, Y, Z) of the cell, the SiC substrate's thickness (t), the count of Si atoms, and the number of solvent molecules. Then the applied three-dimensional periodic boundary conditions were set along the X, Y, and Z directions, the number of simulation steps at 2,000,000 and the time step to 0.25 fs, NVT ensemble, and SiC (Si/C/O/H/N/S) force field were employed to carry out the simulation process. The temperature is regulated by a Berendsen thermostat possessing a damping coefficient of 5 fs, and the overall duration of the simulation is 500 Ps. All simulation processes were completed using the AMSjobs [40,47,48] calculation software ReaxFF module, the software version number is AMSjobs 2021.106. The influence of temperature on reaction kinetics implies that maintaining the desired kinetic behavior at the initial temperature can be achieved by increasing the temperature of the reaction system. To clearly illustrate the disparity in reaction rates between the sic surface and various solvents, three distinct temperatures were chosen to manifest the contrasting chemical reactivity of solvents on the sic surface. To study the effect of temperature, three temperature conditions, 300 K, 1000 K, and 2000 K, were adopted.

Table 2
CMP process parameters.

Process parameters	Conditions
Substrate	C/Si -face SiC, 2-inch
Polisher	ZDHP-30
Pad	Self-sharpening FAP
Pressure	17.2 KPa
Slurry flow rate	20 ml/min
Platen/carrier speed	20/50 r/min
Polishing time	10 min

2.3. Materials and polishing experimental methods

In the polishing experiment, a colloidal silica polishing slurry with a particle size of 120 nm and a concentration of 20 wt% (Shanghai Yingzhi Grinding Co., Ltd.) was used to pre-polish the C side and Si face of the 2-inch SiC single wafer (Jiangyin Lanke Crystal Materials Co., Ltd., China) at 0.8 psi pressure for 30 min to make the wafer surface state consistent. Deionized water, methanol (CH_3OH, CAS no.67-56-1, ≥99.5 %), and ethanol (C_2H_5OH, CAS no.64-17-5, ≥99.7 %) were prayed onto the fixed abrasive pad to polish the *C*-side and Si-side of the SiC wafer, respectively. The CMP polishing investigations were executed upon an ultra-precise annular polishing apparatus (ZDHP-30, Zhibang, China) utilizing a hydrophilic self-sharpening adhesive abrasion surface measuring 300 mm in diameter, which had been furnished by Nanjing University of Aeronautics and Astronautics. Each chemical-mechanical planarization (CMP) experiment was conducted at an applied pressure of 17.2 Kpa for a duration of 10 min, with the solvent flow rate set at 20 mL per minute, and the speed of the platen and carrier set at 20 and 50 revolutions per minute, respectively (Table 2). Each CMP experiment was replicated three times. Each CMP experiment was repeated 3 times. The experiment of solvothermal corrosion of SiC was carried out in a 100 ml capacity hydrothermal vessel. SiC particles were placed in three vessels filled with 30 ml of deionized water, CH_3OH, and C_2H_5OH solvents respectively. The filled vessels were sealed and placed in an oven at 120 °C for 120 h. The reaction products were analyzed for XPS analysis.

The SiC wafer masses pre- and post-polishing were assessed utilizing an OHAUS precision analytical balance, boasting a degree of accuracy to within 0.01 mg. Subsequent to this, MRR values were computed via appropriate methodologies [49].:

$$MRR = \Delta m/\rho\pi r^2 t \quad (1)$$

Concerning the given equation, the variable Δm indicates the quantity of mass that has been lost. The symbol ρ corresponds to the specific density of SiC which has a distinct value of 3.2 g/cm^3. The variable r represents the radius of the SiC substrate, while t represents the duration of the polishing process.

An Atomic Force Microscope (AFM, Nanosurf Isostage 300, Nasser Nanotechnology Co., Ltd.) was employed to gauge the surface morphology and roughness of the meticulously polished Silicon Carbide (SiC). The evaluated test region amounted to 10 μm × 10 μm, with each specimen subject to measurement at three distinct points. The elemental composition of the solvent-etched silicon carbide wafer was scrutinized using X-ray photoelectron spectroscopy (Thermo Scientific K-Alpha) with Al Kα (hv = 1486.6 eV) radiation.

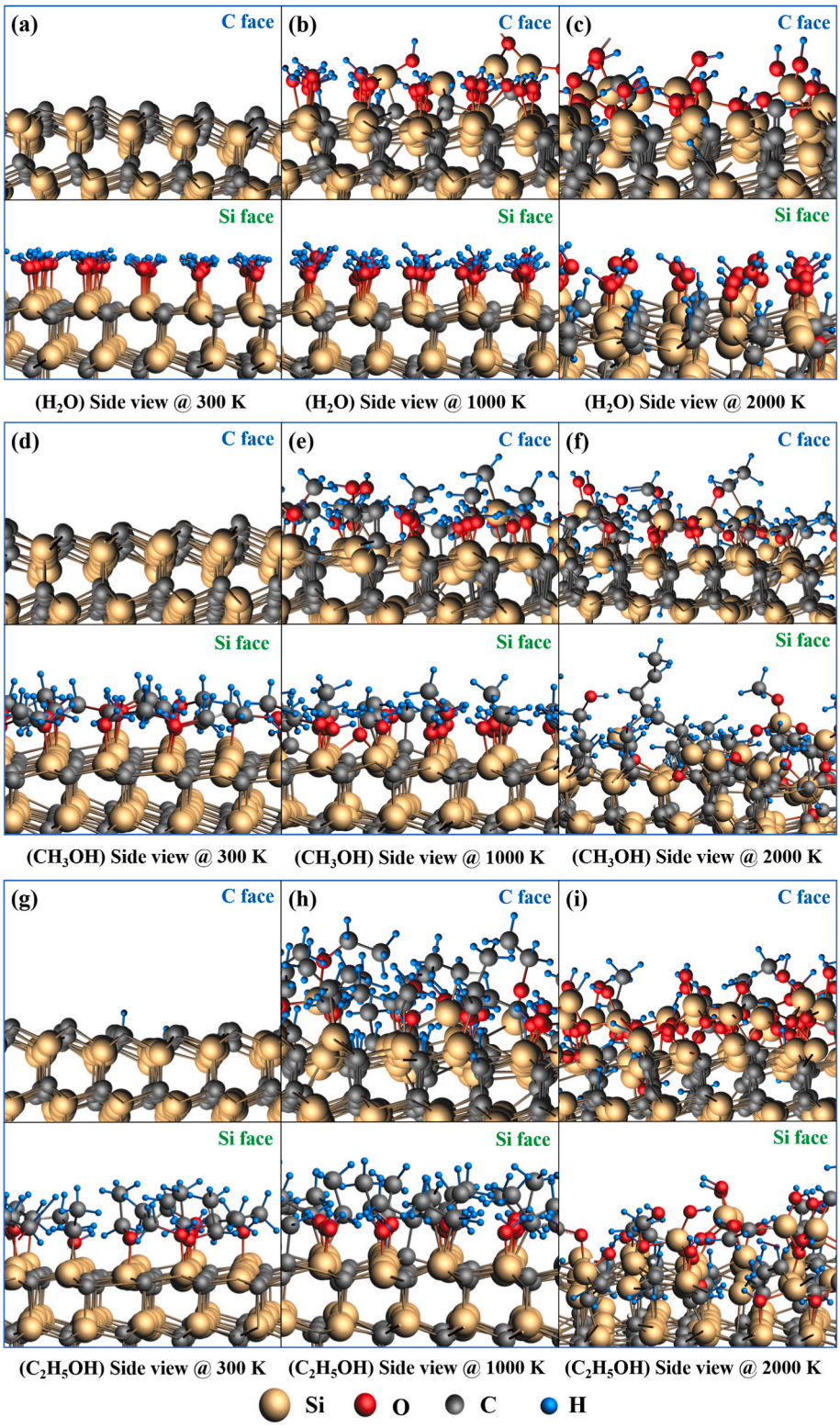

Fig. 2. Side views of the SiC surfaces reacting with (a) H_2O, (b) CH_3OH, and (c) C_2H_5OH after 500 Ps simulations at 300 K, 1000 K, and 2000 K.

3. Results and discussion

3.1. Surface structures of SiC after being exposed to H_2O, CH_3OH, C_2H_5OH

Sorensen and Voter [50] adopted the assumption of the simple harmonic transition state theory and successfully achieved temperature-accelerated reaction kinetics. The time required for ReaxFF molecular dynamics simulation can be significantly shortened by raising the simulation reaction temperature while the reaction kinetic remains undisturbed. This approach is also employed in our simulation. The simulation is conducted at 300 K, 1000 K, and 2000 K for 500 Ps. After the SiC surface is exposed to H_2O, CH_3OH, or C_2H_5OH for 500 Ps, the potential energy of the system tends to be stabilized, and the chemical

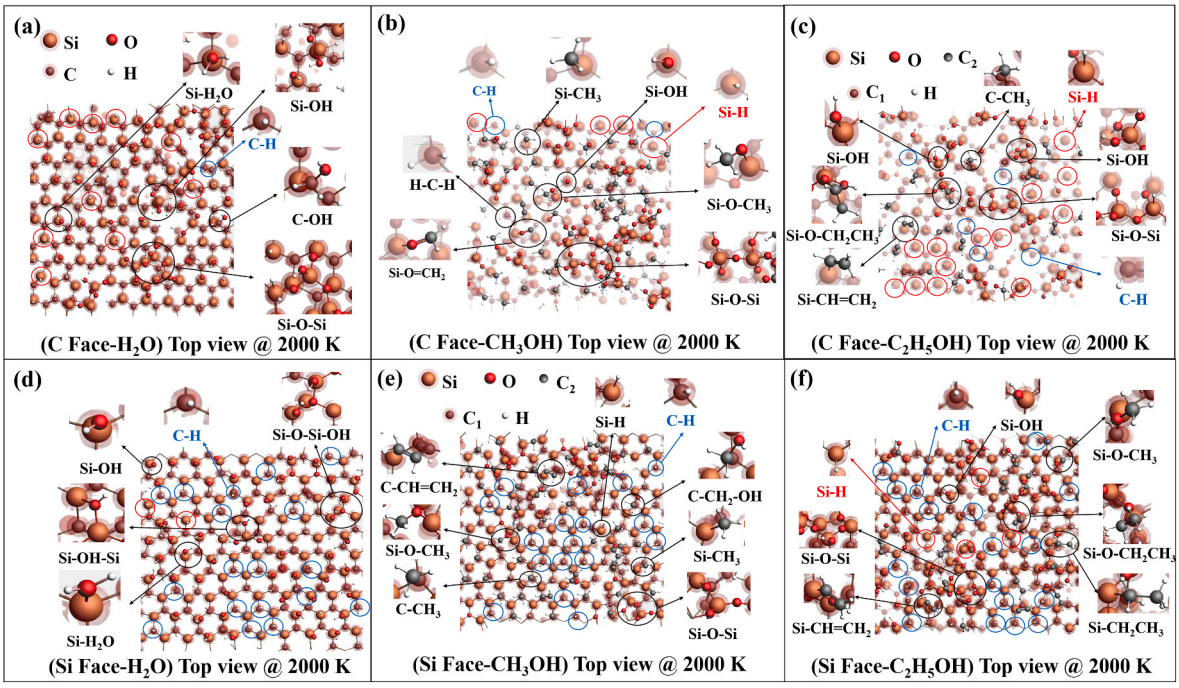

Fig. 3. Top views of the SiC surfaces after 500 Ps simulations at 2000 K.

reaction on the SiC surface reaches a steady state.

Fig. 2 presents a side view of the solvent reacting with the SiC surface at 300 K, 1000 K, and 2000 K while Fig. 3 is a top view of the solvent reacting with the SiC surface at 2000 K.

From the side view of SiC at 300 K (Fig. 2(a)), H_2O molecules are preferentially adsorbed on the Si face of SiC and form a uniform monolayer, but no H_2O is found on the C face at all. Both CH_3OH and C_2H_5OH are also preferentially adsorbed on the Si face of SiC substrates with a crowded surface alcohol molecule layer. Compared to the existence of H_2O, there are a few alcohol species that are adsorbed on the C face, a trace amount of indication about alcohol reactivity towards Si face of SiC. At 300 K, all three solvent molecules mainly physically adsorb on the surface of SiC.

As the temperature rises to 1000 K, the unreactive C face of SiC at 300 K suddenly becomes very reactive with H_2O as shown in Fig. 2(b), some *Si*–C bonds are broken, and their reaction products are seen leaving the SiC surface. Surprisingly physically adsorbed H_2O layer on the Si face of SiC keeps its well-arranged structure intact. These observations provide evidence about the more reactive nature of the C face *versus* Si face even though H_2O molecules demonstrate strong affiliation with the Si face of SiC at 300 K. For alcohol molecules at 1000 K, chemical reactions are similar to H_2O, which take place on C face, but not on Si face. One does have to take notice that the adsorbed water structure on the Si face at 1000 K is almost the same as that at 300 K (Fig. 2(b)), but the alkyl groups of alcohol molecules start moving away from the SiC surface (Fig. 2(e) and (h)), driving the OH functional group in close contact with Si atoms on the surface that leads to visibly longer *Si*–C bond on SiC.

At 2000 K, many more *Si*–C bonds on the C face of SiC are broken as shown in Fig. 2(c), and the surface Si and C atoms react with H_2O to form Si–OH and *C*–OH species. On the Si face, surface Si–OH and *Si*–H species can be clearly identified though the Si atom on the Si face is still strongly bonded to the bulk SiC structure as the SiC crystal structure remains intact.

For alcohol molecules on the C face, varieties of chemical species are accumulated on the surface, and its detailed analysis will be presented in the following paragraph. The same can be said for reaction products on the Si face. To evaluate the detailed surface species on the surface of SiC, the top view of surface species on SiC after being exposed to water and alcohols at 2000 K is presented in Fig. 3. It can be seen that the SiC surface in aqueous solvent covers with terminal H, OH, H_2O. On the other hand, in addition to the terminal H, OH, a large amount of alkyl and alkoxy groups are found on the surface of SiC in alcohol solvents, and there is also a small amount of surface vinyl species.

The top view of the C face of SiC in the H_2O solvent system at 2000 K reveals that there are *Si*–H, *Si*–H_2, *C*–H, adjacent SiOH–SiOH, *Si*–*O*–Si reaction products and physically adsorbed H_2O. Those hydride *Si*–H, *Si*–H_2, and *C*–H species are chemical reaction results of surface Si or C with H_2O. *Si*–H is also observed when the Si wafer is polished with an aqueous polishing slurry, suggesting that the Si atoms on the surface of the Si wafer and SiC experience similar chemical reaction processes though their reactivity differ significantly from each other. Similar reaction products are found on the Si face of SiC substrates (Fig. 3) except no physically adsorbed H_2O is found, consistent with the hydrophobic nature of the Si-face. What is intriguing is that no matter whether the surface is Si or C on SiC substrates, the surface terminal species after being exposed to H_2O are the same, in the form of Si–OH and Si–OH_2 instead of *C*–OH and *C*–H_2O, an indication that Si–OH and Si–OH_2 are more thermodynamically stable than *C*–OH and *C*–H_2O on SiC surface under our simulated conditions. It is worth noting that the Si face is mainly H-terminated in the form of *C*–H (Fig. 3 blue circle mark), while the *C*-surface is mainly H-terminated in the form of *Si*–H (Fig. 3 red circle mark). We speculate that on the Si face, when the *Si*–C bond is broken by reacting with H–OH, Si forms thermodynamically more stable Si–OH, the resulting H will bond with C to form *C*–H, *vice versa*.

Different from H_2O solvent, in the CH_3OH, C_2H_5OH solvent system at 2000 K, it can be seen from the top view of Si and C face of SiC that in addition to the presence of H, OH, CH_3OH, C_2H_5OH, a variety of C_nH_{2n+1} (n = 1, 2), –OCH_3, –CH_2OH, –O═CH_2, –CH═CH_2 groups can be identified on the reacted surface. Those alkyl (-CH_3, –CH_2CH_3) and alkenyl (-CH═CH_2) on the C face SiC are directly bonded to surface Si atoms to form organo-silicon species.

On the SiC surface in all three solvents, H exists in two forms on C face: monohydride *C*–H, *Si*–H and dihydride CH_2, *Si*–H_2 (Fig. 3). On the Si face, however, only monohydride species C(Si)–H can be observed. In addition, oxygen atoms can penetrate deep into the SiC substrate to

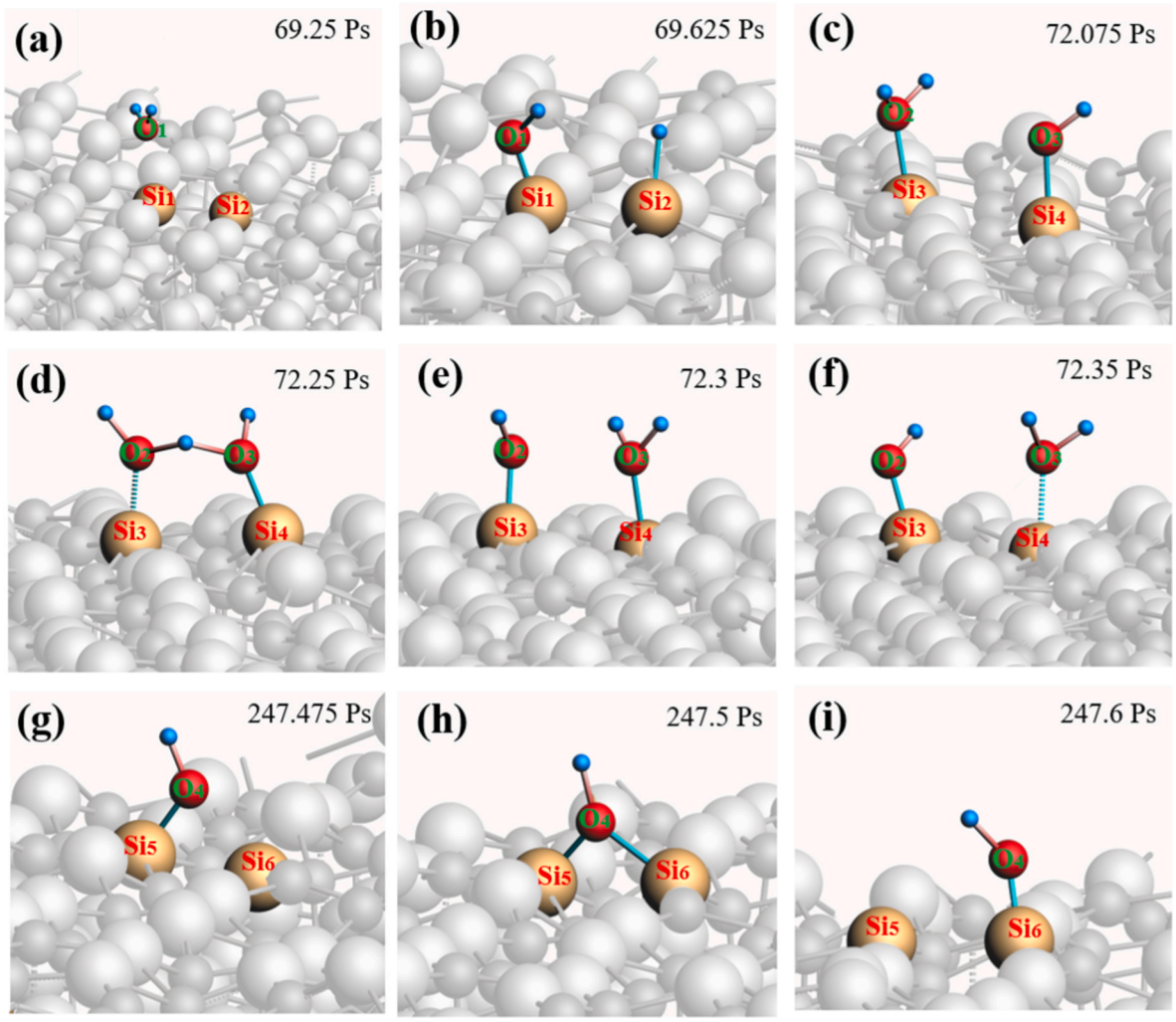

Fig. 4. Interaction between H_2O and SiC substrate surfaces. (a) H_2O close to the SiC surface, (b) H_2O dissociates on the SiC surface to form Si–OH and *Si*–H, (c) H_2O and ortho-Si-OH adsorbed on Si atoms on the SiC surface, (d) formation of OH–H–OH intermediates at the interface, (e) H_2O exchanges with ortho-Si-OH, (f) newly formed H_2O desorption on the SiC surface, (g) adsorbed OH close to the ortho-Si atom, (h) Si–OH–Si intermediates, (i) OH is transferred to another Si atom.

break *Si*–C bonds and form *Si*–*O*–Si, suggesting that as reaction temperature rises, oxygen atoms diffuse deeper into the bulk of the SiC surface to form *Si*–*O*–Si species (Fig. 2).

3.2. Surface reaction mechanisms of H_2O, CH_3OH, C_2H_5OH on SiC substrates

3.2.1. Surface reaction of H_2O on SiC substrates

Fig. 4 graphically depicts the detailed surface reaction process of SiC with H_2O solvent, where the reacting molecules are marked and highlighted to clearly track the evolution of reaction steps. The process starts with a H_2O (marked as O_1) molecule diffusing close to the SiC surface (Fig. 4(a)), followed by breaking the H–OH bond to form a Si–OH species with surface Si (marked as Si_1) (Fig. 4(b)) and *Si*–H with another neighboring Si (marked as Si_2). The Si (marked as Si_4) adjacent to the formed Si–OH (marked as Si_3, O_3) may become chemically more reactive, and initiate strong chemical adsorption with an H_2O molecule (marked as O_2) from the liquid phase. The bond strength of one OH group in chemical-sorbed Si–HO–H is weakened, and bridges with neighboring Si–OH to form HO–Si–OH–Si intermediate as depicted in Fig. 4(d), this meta-stable intermediate is then deprotonated and transferred at the interface through the water phase Fig. 4(e), resulting in that Si–OH_2 turns into Si–OH, and the adjacent Si–OH becoming Si–OH_2, of the OH_2 that eventually leaves the adjacent Si (Fig. 4(f)), a switching identity of two adjacent Si_3 and Si_4 atoms in presence of liquid water. In addition, the newly formed Si–OH is also found to be capable of hopping between two neighboring Si as illustrated in Fig. 4(g)–(i)). The OH group bonded with one Si (marked as Si_5) atom first approaches the adjacent Si_6 atom to form Si–OH–Si, and then transfers to the adjacent Si atom via the Si–OH–Si intermediate (Fig. 4(g)–(i)), completing a cycle of OH group hopping.

3.2.2. Surface reaction of alcohols on SiC substrates

The reaction process of CH_3OH and C_2H_5OH on SiC surfaces are similar, therefore only the reaction path of CH_3OH solvent on SiC is presented in Fig. 5. For a better visual illustration, CH_3OH molecules are marked and highlighted in these processes.

Different from the adsorption and dissociation of H_2O on the surface of SiC, there are two reaction pathways for CH_3OH on the SiC surface. One is that when CH_3OH diffuses towards the SiC surface, the O atom (marked as O_1) in CH_3OH is selectively adsorbed on a Si (marked as Si_1) atom (Fig. 5(a)) and then decomposes on the Si face of SiC to form Si–OH and an organosilicon species *Si*–CH_3 (Fig. 5(b)), similarly to what happened with H_2O. The other pathway is that when CH_3OH approaches the Si face of SiC, its O atom of the CH_3OH directly bonds to the surface Si to form *Si*–$OHCH_3$. The adjacent Si–OH captures the H in *Si*–$OHCH_3$ intermediates to generate *Si*–H_2O and *Si*–OCH_3, and the H_2O molecule of the formed *Si*–H_2O later desorbs and leaves from the Si face (Fig. 5(e)).

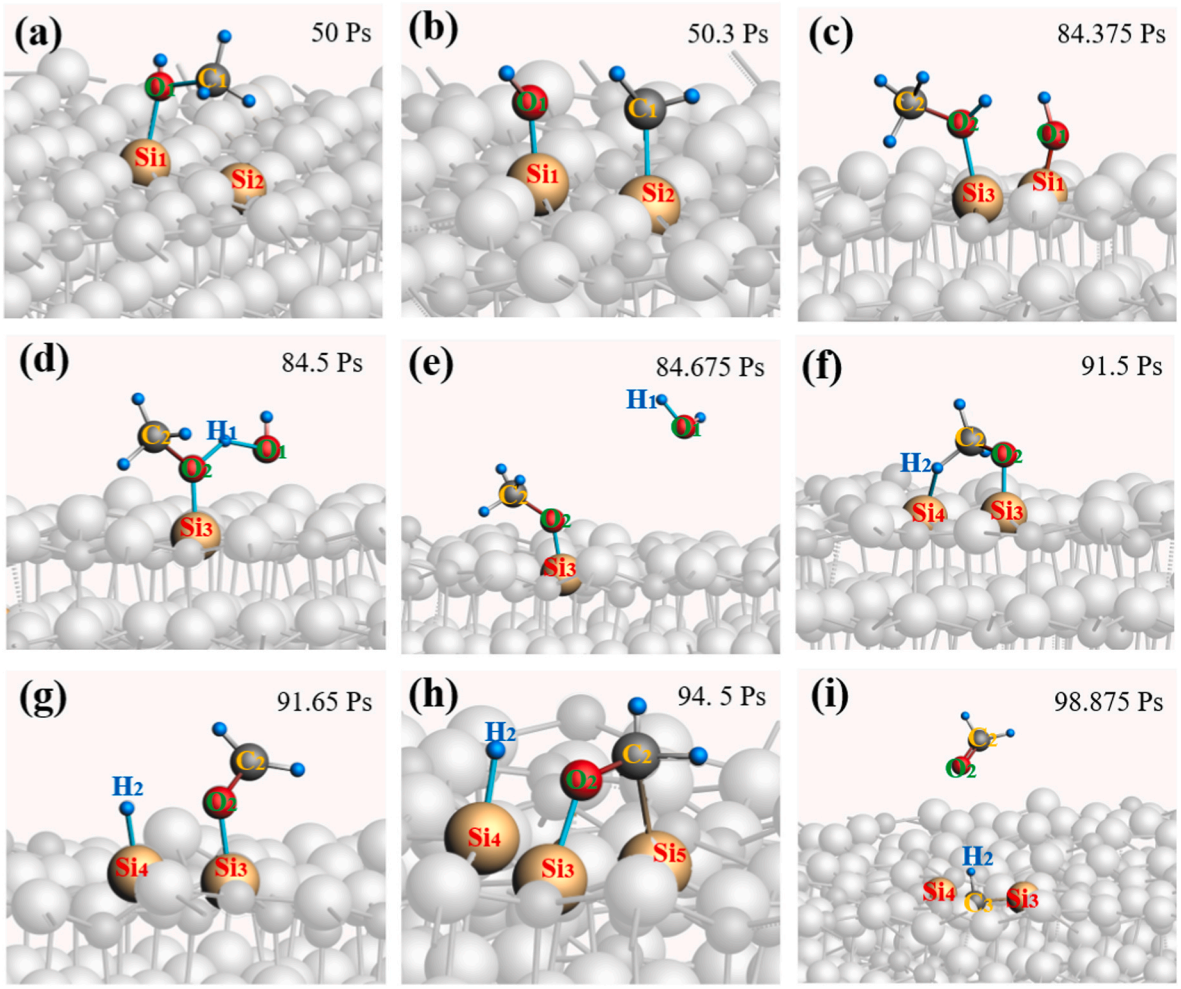

Fig. 5. Reaction between CH_3OH and SiC substrate surface: Yellow is Si atom, red is O atom, gray is C atom, and white is H atom. (a) CH_3OH adsorbed on the surface of SiC, (b) CH_3OH dissociated on the surface of SiC to form Si–OH and *Si*–CH_3, (c) CH_3OH and ortho-Si-OH adsorbed on Si atoms on the surface of SiC, (d) formation of OH–H–OCH_3 intermediate at the interface, (e) H_2O formed by OH seizure of H desorption on the surface of SiC, (f) Si adsorption of H protons in *Si*–OCH_3, (g) Si seizure of H protons in *Si*–OCH_3, (h) *Si*–OCH_2–Si intermediates, (i) H is transferred to the ortho-C atom, CH_2O is detached from the SiC surface. (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

Therefore, two surface silicon species, *Si*–CH_3 and *Si*–OCH_3, are formed when CH_3OH is exposed to the SiC surface, which is consistent with the simulation results by Silvestrelli using first principles [51], in which the *Si*–$OHCH_3$ is a chemically kinetically favored reaction product while –OH and –CH_3 species is thermos dynamically preferred on SiC surface. Notably, both surface reaction paths are reversible. An increase in reaction temperature often leads to CH_3OH reaction switching from the *Si*–$OHCH_3$ to *Si*–CH_3 formation path.

The CH_3O–Si species goes through further reaction as follows. An adjacent Si atom bonds with one H atom of the CH_3–*O*–Si to form *Si*–H and CH_2–*O*–Si surface species (Fig. 5(f)–(g)), and the CH_2–*O*–Si goes through a *Si*–*O*–CH_2–Si intermediate state (Fig. 5(h)), eventually breaking the *Si*–C and *Si*–O bonds to form a CH_2O molecule to leave the surface (Fig. 5(i)).

Ethanol CH_2CH_3OH displays quite a different reaction path on the SiC surface. On the Si face of SiC, CH_2CH_3OH reacts with surface Si to form Si–OH and *Si*–CH_2CH_3. Of the later *Si*–CH_2CH_3, an uncoordinated ortho-Si bonds to the H of the terminal C of *Si*–CH_2CH_3 on the surface (Fig. 6(a)) to form *Si*–H and *Si*–CH_2CH_2 intermediates (Fig. 6(b)), eventually the intermediate *Si*–CH_2CH_2 releases a $CH_2{=}CH_2$ molecule, by breaking *Si*–C bond from SiC surface (Fig. 6(c)); The other is that *Si*–CH_2CH_3 captures H in *Si*–H (Fig. 6(d)), breaks the *Si*–C bond (Fig. 6 (e)), and release a C_2H_6 molecule from the surface of silicon carbide (Fig. 6(f)). These two reaction paths facilitate the transfer process of H atoms on the SiC surface.

3.2.3. *Chemical reactivity of solvent molecules on SiC surface*

The Chemical reactivity of water and alcohols on the SiC surface are compared by plotting the number of unreacted solvent molecules, the number of OH and H adsorbed on the surface, the number of O atoms that form *Si*–*O*–Si bonds, and the number of Si atoms and the number of C atoms that are detached from the reacted SiC surface versus reaction time in Fig. 7.

The unreacted CH_3OH and C_2H_5OH molecules are much fewer than H_2O under the same reaction conditions over time (Fig. 7(a)). At 1000 K, no matter whether it is alcohols or H_2O, the consumption of solvent molecules seems to reach a steady state after the first 100 Ps while a rapid alcohol molecule consumption keeps about the same rate through the whole reaction process at 2000 K, suggesting that at higher reaction temperatures, reaction products can quickly leave the reaction surface to expose fresh SiC for reaction to continue, and the kinetics of reaction products leaving reaction interface can keep up with the chemical reaction rate. While at lower temperatures, the reaction kinetics may be hindered by the desorption of reaction products from the reacted SiC surface. It is worth noting that even at 2000 K, the chemical reactivity of H_2O with SiC surface remains slow, and seems to stop after the first 100 Ps, similar to what happened at 1000 K to alcohol molecules, one can speculate that the reaction products of water with SiC at 2000 K are

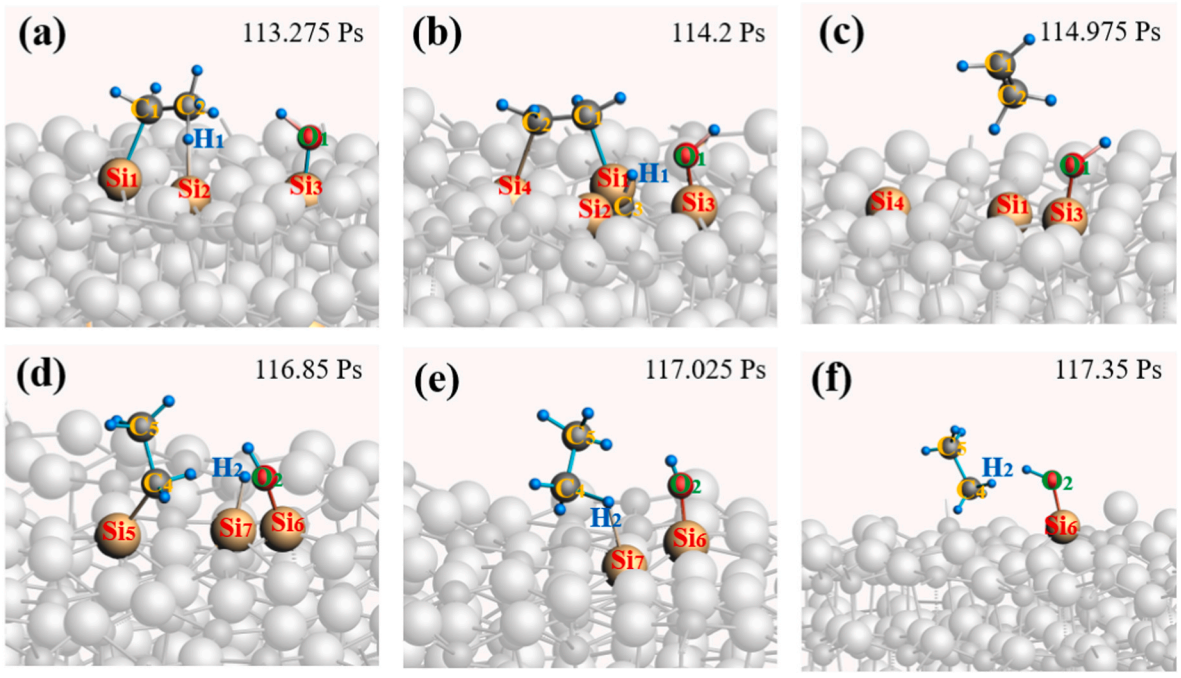

Fig. 6. The reaction paths of ethanol on SiC.

thermodynamically stable on the reacted surface, no enough kinetic driving force to regenerate the reactive surface for reaction to continue.

The OH groups on the SiC surface present the most intriguing information as shown in Fig. 7(b). At 1000 K for the first 100 Ps, the number of surface OH decreases as $CH_3OH > C_2H_5OH > H_2O$. The number of OH groups slowly increases over reaction time for H_2O, reaches a maximum after 100 Ps, and becomes almost constant thereafter. For C_2H_5OH molecules on SiC at 1000 K, the number of produced surface OH groups steadily keep increase over time. The number of surface OH for CH_3OH at 1000 K rapidly increases, reaches a maximum at about 50 Ps and remains flat thereafter. At 2000 K, the number of OH on the surface goes $H_2O > C_2H_5OH > CH_3OH$, a complete reverse of orders compared to that at 1000 K, the highest number of surface OH for H_2O at 2000 K might be best interpreted as the reason for low reactivity of H_2O at 2000 K because of its high stability to remain on the SiC surface. The number of OH groups steadily increases with reaction time for H_2O while the OH groups for alcohols begin to steady drop after a quick rise for the first 50 Ps, especially for methanol, a clear indication of OH consumption as reactants for the following reaction on the SiC surface as reaction time proceeds.

The number of H species on the SiC surface shows that C_2H_5OH produces the most H species compared to CH_3OH and H_2O, and H_2O generates slightly more H species than methanol. There is no oxygen in the form of *Si*–*O*–Si, and no Si and C, which break their *Si*–C bonds in the bulk structure of SiC and form chemical bonds with solvent molecules from the balk SiC structure, to be found on the SiC surface at 1000 K. However, the total number of surface O atoms that form *Si*–*O*–Si bonds on the SiC surface at 2000 K increases with reaction time in Fig. 7(d), and follows the order of $C_2H_5OH > CH_3OH > H_2O$. The surface oxygen in the form of *Si*–*O*–Si must come from oxygen-containing solvent molecules since there is no oxygen in SiC substrates, and the number of oxygens in the form of *Si*–*O*–Si can be a good indicator to judge the reactivity of solvent molecules on SiC. The same can be said about the number of Si species that demonstrate the same pattern with reaction time. There is no C in H_2O molecules, so it is no surprise no C species found on the SiC surface reacted with H_2O. One can hardly differentiate CH_3OH from C_2H_5OH on the number of C at 2000 K.

Fig. 7(a) suggests that alcohols are very reactive on the SiC surface since they are consumed the most at any temperature and reaction time, and the surface OH that is generated by methanol reacting with SiC is more reactive at 2000 K, and can be quickly consumed by the following reaction as reactants. However, ethanol generates more surface hydrides than methanol and water at 2000 K.

3.3. Oxidation reaction mechanisms of solvents on SiC surface

3.3.1. Oxidation of SiC surface by H_2O

The chemical reaction of SiC with solvent molecules is investigated by tracking the reaction steps of the marked Si, C, and H_2O molecules as depicted in Fig. 8. After the initial stage of reactions, the detailed reaction process is shown in Fig. 8(a-i). First, $3H_2O$ molecules react with surface SiC to form 3 Si–OH and 3 *C*–H as depicted in Fig. 8(a), and each Si–OH is surrounded to three *C*–H, so is each *C*–H to three Si–OH. One of the newly formed *C*–H leads to Si(1)-C bond breaking, loosing a *C*–H in the process (Fig. 2(b)). The Si(1)-OH that breaks bonding with *C*–H is activated thereafter, and bonds with neighboring Si(3)-OH to form OH–Si(1)–OH–Si(3) surface species (Fig. 2(c)). Si(1) follows to break its other *Si*–C bond, and bonds with another neighboring Si–OH surface species to form HC-Si(1)=$(OH)_2$–Si (Fig. 8(d)). In Fig. 8(e), Si(1) breaks down its last *Si*–C, to form three coordinated silicon oxy-oxide species. One by one, the three OH groups in surface Si(1) $(OH)_3$ loose their hydrogen to adjacent Si–OH to form *Si*–HOH and Si(1)-*O*-Si (Fig. 8(f-i)). The H_2O desorbs from the SiC surface to restore the surface Si to its initial unreacted state, and leaves the SiC surface partially covered with Si(1)-*O*-Si to complete the SiC surface oxidation process. In a brief summary, H_2O first reacts with the SiC surface to form surface Si–OH and *C*–H. Of the formed *C*–H, OH–Si–C bond is being activated and weakened, and eventually is broken to form Si–OH, leading to *C*–Si-$(OH)_3$ formation. The formed Si-$(OH)_3$ facilitates the last *Si*–C bond to be broken. All three H in Si–OH are readily taken away one by one by its adjacent Si-OHs to form *Si*–*O*–Si and *Si*–HOH. The later *Si*–HOH looses its H_2O, and Si restores to its initial state.

3.3.2. Oxidation of SiC surface by alcohols

The surface species on SiC after reaction with alcohols and gas phase products are identified and presented in Fig. 9. Comparing the surface

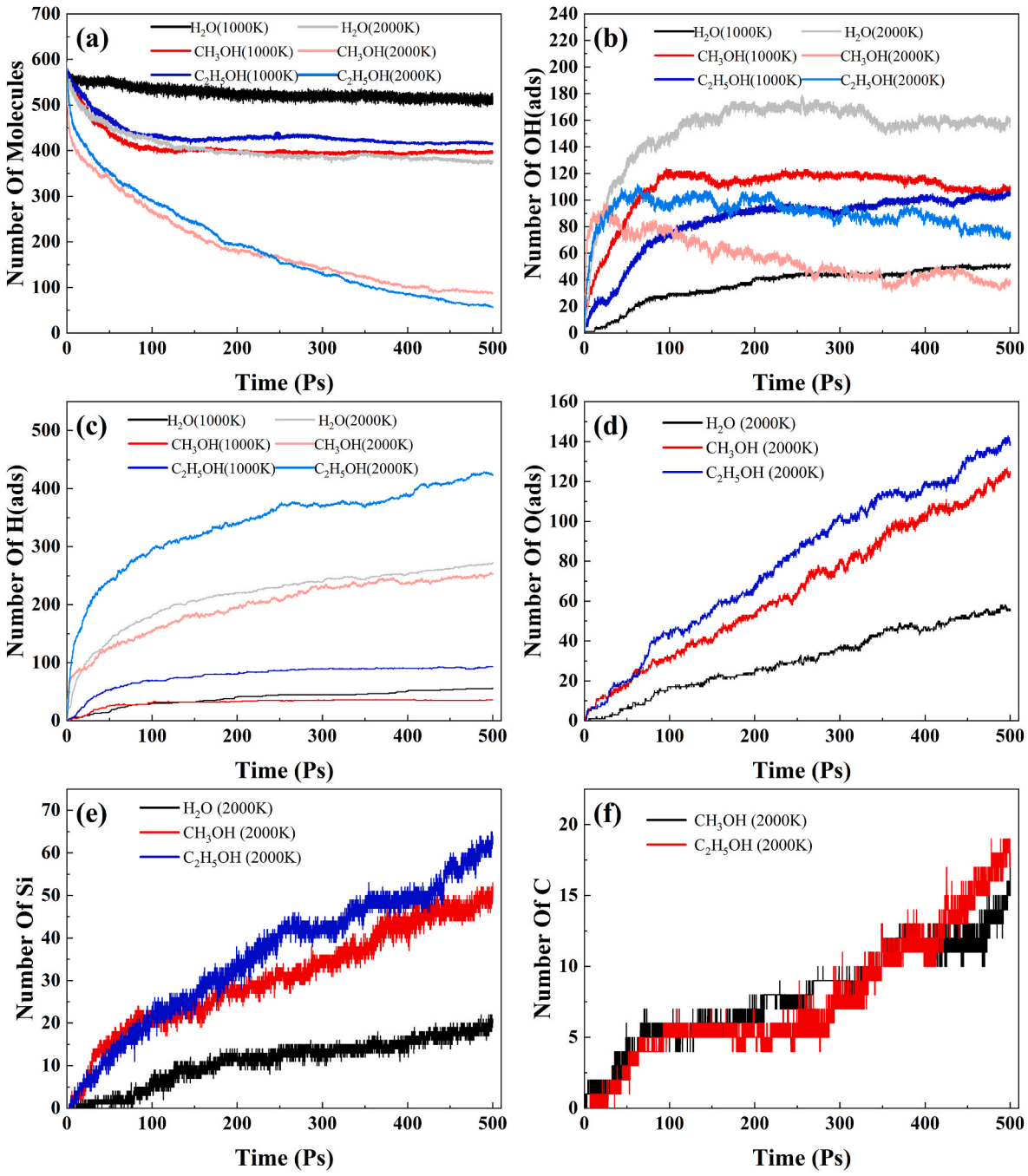

Fig. 7. The number of (a) solvent molecules, (b) the number of SiC–OH adsorption on the surface of SiC, (c) the number of SiC–H adsorbed on the surface of SiC, (d) the number of O atoms forming *Si–O*–Si bonds, (e) the number of Si atoms off-surface, (f) the number of C atoms off-surface during the simulation of interaction with SiC at 1000 K and 2000 K temperatures.

and gas phase species of methanol reaction to that of ethanol at 2000 K, there are many more surface species in absolute numbers on SiC after reacting with methanol than those with ethanol (Fig. 9(a) vs Fig. 9(c)). On the other hand, the opposite is true for the gas phase products in absolute numbers (Fig. 9(b) vs Fig. 9(d)). These numbers clearly illustrate that ethanol is much more chemically reactive than methanol on SiC. For the methanol-SiC reaction at 2000 K, the main surface species are $–CH_3$, $–CH_2$, $–COH_3$ ($-CH_2OH$) with small amounts of $–OCH_3$ and –CH; and the gas phase products are dominated by H_2O, CH_4, and small amount of CH_2O and CH_4O_2(CH_3OOH). For ethanol-SiC reaction at 2000 K, the surface species are very few, mostly $CH_5C(CH_3CH_2)$ and $CH_3C(CH_2–CH)$, and abundant amount of gas phase products are found to be C_2H_4, H_2O and small amount of C_2H_4O (CH_3COH) and C_2H_6. The dominant gas phase products in ethanol-SiC reaction point to the reactive nature of the ethanol-SiC surface that drives out any intermediates.

Following the same process as used for H_2O, the main oxidation reaction process of SiC by methanol is tracked and presented in Fig. 10. First as CH_3OH molecule diffuses to the SiC surface, H bonds with the surface C to form C3–H (Fig. 10(a)). Indeed, one can find many more *C*–H species by carefully examining the unhighlighted background. The formation of *C*–H leads to the surface *Si*–C bond reconstruction, and the Si atom is pulled up and exposed to the surface (Si_1, Si_2, Si_3, and Si_4 in Fig. 10(b)). Should there be no other surface species except Si, the adjacent Si would react, in turn, with activated CH_3 group of the

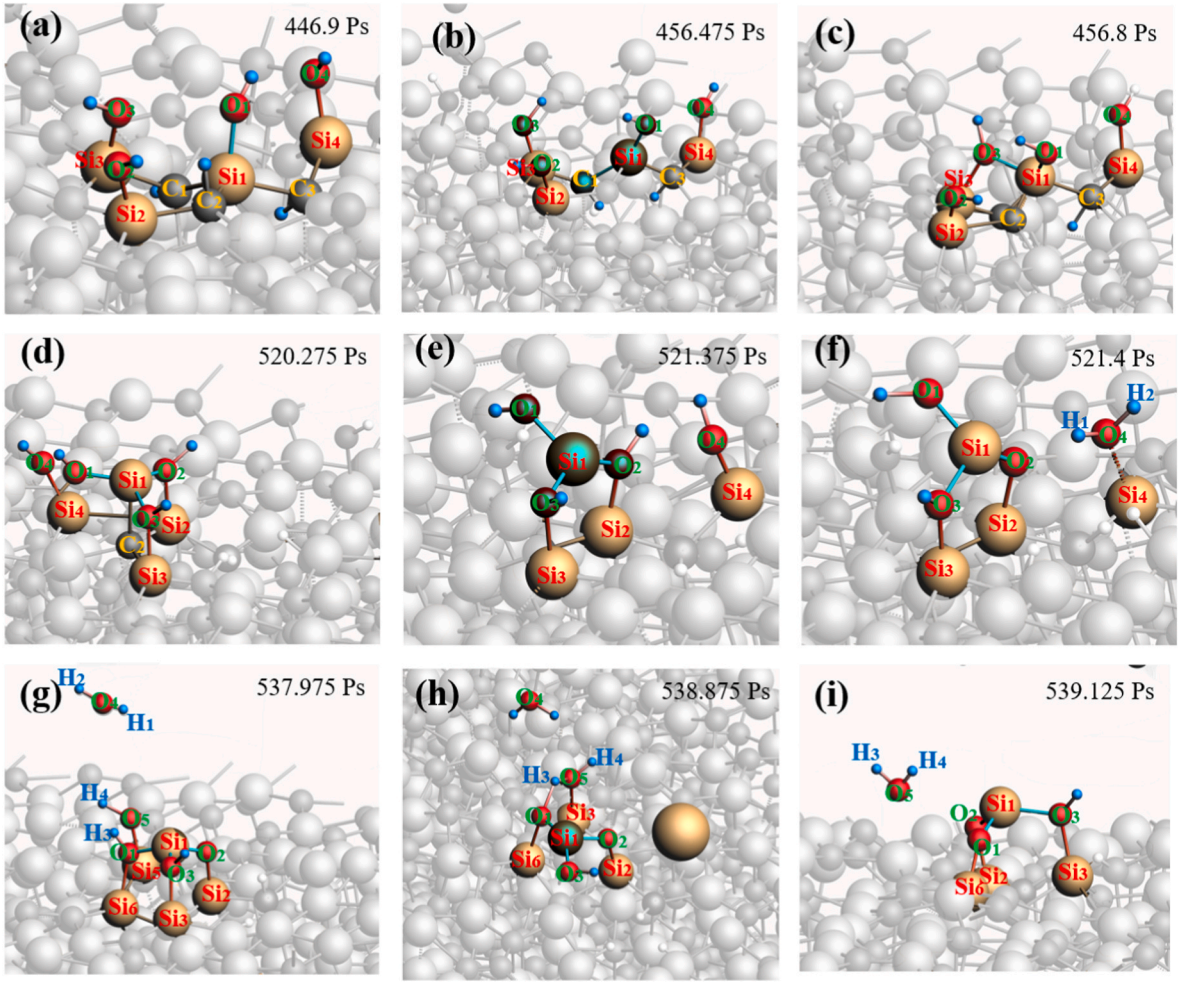

Fig. 8. The oxidation mechanism on the surface of SiC in H_2O. yellow is Si atom, red is O atom, gray is C atom, white is H atom (a) OH adsorbed on the surface of SiC, (b) surface *Si*–C bond breakage, (c) Si–OH–Si intermediate formation, (d) OH transfer on the SiC surface, (e) OH transfer to intermediate ortho, (f) OH seizure of H protons in Si–OH–OSi, (g) H_2O generated by Si leaves the surface, (h) OH continues to seize H protons in Si–OH–OSi, and (i) H_2O leaves the SiC surface. (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

intermediate to form *Si*–CH_3 and Si–OH (Si_2–CH_3)$_2$ in Fig. 10(c)). Should there be Si–OH near the newly formed *Si*–CH_3 intermediate as shown Si_2–CH_3 and Si(1)-OH in Fig. 10(d), Si(1) in Si(1)-OH initiates bonding with H in *Si*–CH_3 to form an intermediate *Si*–H_2C–H–Si–OH (Fig. 10(e)). Eventually, the *Si*–CH_3 snaps away the H from Si–OH to form *Si*–*O*–Si and CH_4 (Fig. 10(f)). The CH_2O substance produced in the previous reaction (Fig. 5(i)) can also form an intermediate of *Si*–OCH_2 with exposed surface Si (Fig. 10(g)), which reacts with the neighboring Si–OH, forming *Si*–*O*–Si bonds and *Si*–OCH_3 (Fig. 10(h)). The resulting *Si*–*O*–CH_3 and a neighboring Si atom form a Si - (*O*–CH_3) – Si intermediate (Fig. 10(i)), followed by *C*–O bond cleavage (Fig. 10(j)), forming *Si*–CH_3 and *Si*–*O*–Si (Fig. 10(k)). Here, three reaction pathways lead to soft surface *Si*–*O*–Si formation after CH_3OH exposure to the surface of SiC: the 1st path is *Si*–CH_3 snaps H from the neighboring Si–OH to produce CH_4 and *Si*–*O*–Si; the 2nd path is the reaction of *Si*–OCH_2 with adjacent Si–OH to produce *Si*–OCH_3; the 3rd path is the *C*–O bond fracture of *Si*–OCH_3, forming *Si*–CH_3 and *Si*–*O*–Si with neighboring Si. These many pathways accelerated soft *Si*–*O*–Si formation, in turn, the MRR of SiC polishing in alcohols.

The main reaction process of ethanol with SiC follows similar reaction mechanisms, and will not be detailed in this manuscript. Combining results from Figs. 9 and 10, the reaction mechanisms for methanol can be outlined as follows:

$$C_nH_{2n+1}OH + SiC \rightarrow C_nH_{2n+1}\text{-HO-SiC} \quad (2)$$

$$C_nH_{2n+1}\text{-HO-SiC} + SiC \rightarrow C_nH_{2n+1}\text{-SiC} + HOSiC \quad (3)$$

$$C_nH_{2n+1}\text{-HO-SiC} + OH\text{–SiC} \rightarrow C_nH_{2n+1}\text{-}O\text{-SiC} + HOH\text{–SiC} \quad (4)$$

$$HOH\text{–SiC} \rightarrow SiC + H_2O \quad (5)$$

$$C_nH_{2n+1}\text{-}O\text{-SiC} + SiC \rightarrow C_nH_{2n}\text{-}O\text{-SiC} + H\text{–SiC} \quad (6)$$

$$C_nH_{2n}\text{-}O\text{-SiC} \rightarrow C_nH_{2n}\text{-O Si=C} + C \quad (7)$$

$$C_nH_{2n}\text{-O} + Si \rightarrow C_nH_{2n}\text{-O} + Si= \quad (8)$$

$$Si\text{–}CH_3 + Si\text{–OH} \rightarrow Si\text{–}O\text{–Si} + CH_4 \quad (9)$$

$$Si\text{–}OCH_2 + Si\text{–OH} \rightarrow Si\text{–}O\text{–Si} \text{ –}O\text{–}CH_3 \quad (10)$$

$$Si\text{–}OCH_3\text{–Si} + Si \rightarrow Si\text{–}O\text{–Si} + Si\text{–}CH_3 \quad (11)$$

The overall reaction of chemical interaction process to produce small molecular chemical products can be summarized as shown in formula (12):

$$2\ C_nH_{2n+1}OH \rightarrow C_nH_{2n+2} + H_2O\ (n = 1, 2) \quad (12)$$

The outlined reaction processes can clearly explain the identified surface *Si*-C_nH_{2n+1},*Si*–*O*–Si, *Si*–*O*-C_nH_{2n+1} and gas phase H_2O, C_nH_{2n}, C_nH_{2n}-O species. One can speculate the following reactions to take place though not major reactions:

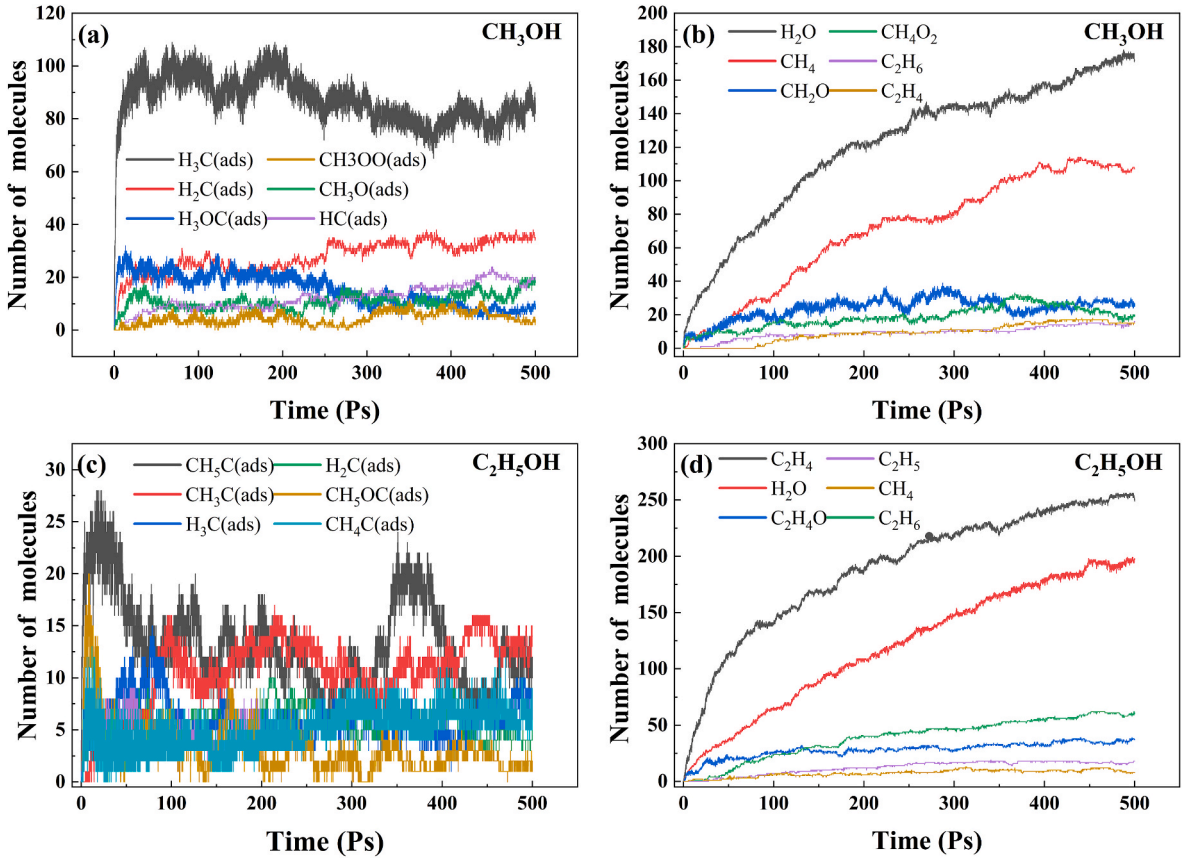

Fig. 9. The main adsorption molecule types, numbers and chemical reaction products in CH_3OH and C_2H_5OH solvent systems at 2000 K. (a) CH_3OH adsorption products on the surface of SiC, (b) small molecule products of CH_3OH interaction on the surface of SiC, (c) adsorption products of C_2H_5OH on the surface of SiC, (d) small molecule products of C_2H_5OH interaction on the surface of SiC.

$$CSi\text{-}C_nH_{2n+1} + SiC \rightarrow CSi\text{-}C_nH_{2n} + H\text{–}SiC \quad (13)$$

$$CSi\text{-}C_nH_{2n} \rightarrow CSi + C_nH_{2n} \quad (14)$$

Which should be able to be found in simulated results should be enough efforts to be made. The three outlined *Si–O–*Si formation pathways are the most persuasive evidence to explain the accelerated MRR of SiC by alcohol solvents.

3.4. Polishing results of SiC on FAP pad

3.4.1. MRR, sa and surface morphology of polished SiC wafers

SiC substrates were polished by deionized water, methanol and ethanol solvents on a fixed abrasive pad. The MRRs of SiC substrates polished are compared in Fig. 11 along with the Sa results. Under the same polishing condition, polishing solvents apply a great influence on the MRR of SiC polishing [39], and exhibit different polishing performance toward to the C face and the Si face of SiC substrates [52]. In polishing experiments carried in this research on the C face (Fig. 11(a)), C_2H_5OH leads to the highest MRR at 43.07 μm/h, followed by H_2O at 41.96 μm/h and CH_3OH at 31.35 μm/h. On the Si face on the other hand (Fig. 11(b)), C_2H_5OH still provides the highest MRR at 82.99 μm/h, followed by CH_3OH at 49.58 μm/h and H_2O at 35.19 μm/h. Basically, no matter on C or Si face, C_2H_5OH displays the highest MRR among all three evaluated solvents, while CH_3OH performs better on material removal of Si face and H_2O polishes faster on C face of SiC, consistent with previous reports that C face is being polished through a reaction mechanism different from that of Si face. It becomes clear that on C face of SiC, aqueous solvent performs equally well as alcohol solvents. On the other hand, on Si face, alcohol solvents polish much more effectively than H_2O, where C_2H_5OH polishes 92.67 % faster than H_2O, and CH_3OH 58.15 %.

Surface roughness of polished SiC substrates, no matter if it is C or Si face, is always lower when polished by alcohols than by water with the highest value at about 12 nm found on C face polished by H_2O. In conventional wisdom, the MRR of CMP polishing goes head-to-head with Sa of the polished surface should the polishing be very mechanically driven. With the same fixed abrasive polishing pad and no free abrasives in polishing solvents, the surface roughness is determined by the only variable, the nature of polishing fluids. For H_2O, surface Sa of polished SiC is always the highest among all these solvents no matter how high or low its MRR, indicating that chemical reactivity and kinetics of H_2O with SiC is lower or slower than those of alcohols. For ethanol with about 97 % MRR improvement, the surface damage Sa on SiC is actually smaller than what H_2O does, therefore one has to contribute the MRR improvement to accelerated chemical reactivity by ethanol with SiC. With this pleasant surprise by using nonaqueous polishing solvents, one can achieve MRR acceleration without sacrificing the process surface quality.

Fig. 12 presents AFM 3D surface topographies of SiC wafers after *C*-face polishing using different solvents. The surface of the *C*-plane wafer polished with DI water (Fig. 12(a)) is covered with deep scratches and pits (be noted the Y axis scale) mixing with shallow scratches, the surface polished with methanol (Fig. 12(b)) and ethanol (Fig. 12(c)) exhibit many more scratches and fewer pits while those surface scratches are much shallower. These observed surface morphologies after polishing strongly support that the surface polishing of the SiC wafer with water is more dominated by mechanical abrasion and cutting, while these actions contributions are counter-compensated when nonaqueous solvents are employed in polishing SiC, resulting in better surface quality. We

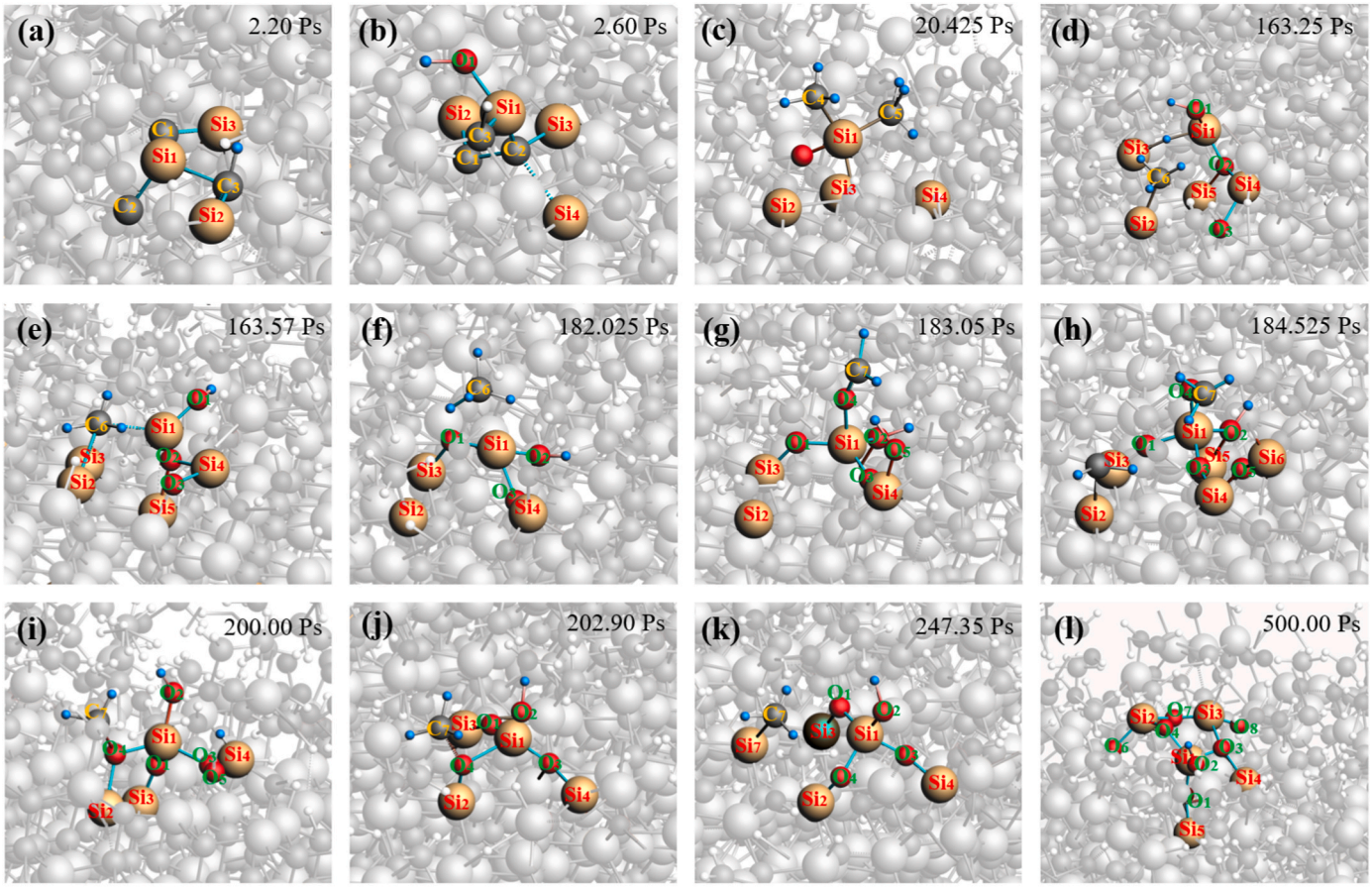

Fig. 10. The oxidation mechanism on the surface of SiC in CH_3OH.

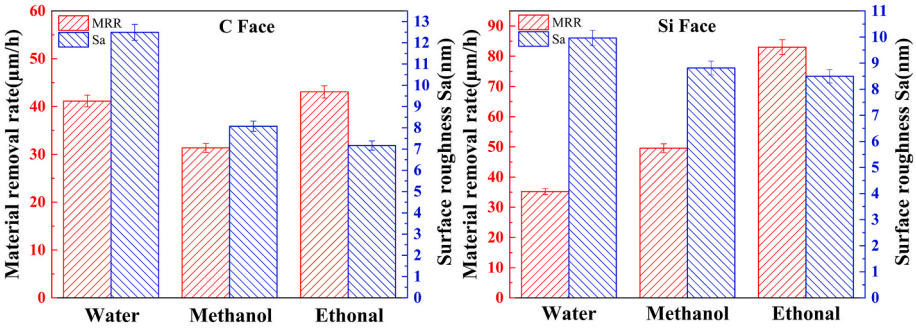

Fig. 11. The MRR and surface roughness Sa of polished SiC C face and Si face with different solvents.

speculate that: the high chemical reactivity and fast kinetics of methanol and ethanol towards to SiC surface allow a top surface softening layer to be formed on the SiC wafer, which reduces the hardness of the wafer surface, makes it easier for the abrading and cutting action between the diamond abrasive grains and the wafer surface, and more plastic removal occurs that reduces surface damage on the surface by abrasives on FAP.

Fig. 13 presents the AFM images and 3D surface topography of the polished Si face of the SiC wafers. Compared to Sa of the polished C face of SiC, the relative Sa improvement on the polished Si face of SiC is smaller in absolute value. However, considering the MRR improvement (>50 %), this Sa improvement becomes more significant. It is noteworthy to mention that, in general, the Sa of the polished Si face is comparatively smaller than that of the C face, which implies that the Si face exhibits a stronger mechanical resistance towards abrasion. In addition, the surface morphology of both the Si and C faces of the wafer that has been polished using H_2O demonstrates similar outcomes. On the other hand, when comparing the H_2O (Fig. 13(a)) polished Si face of the SiC wafer with the Si face that has been polished using methanol and ethanol (Fig. 13(b) and (c)), it is observed that the former exhibits more surface protrusions. This is attributed to the lack of chemical interaction between the solvent and the wafer surface, the difficulty in forming a surface softening layer, and the result of rigid removal between the wafer surface and the mechanical friction of diamond abrasive grains. In addition, on the wafer surface polished by the same solvent, the H_2O polished Si face (Fig. 13(a)) has better surface quality than the C face (Fig. 12(a)). However, the surface quality of the *C*-face of the wafer treated with alcohol polishing is better than that of the Si-side. According to Figs. 12 and 13, there exists a discrepancy in the Sa data acquired via AFM and 3D morphology measurements of polished SiC

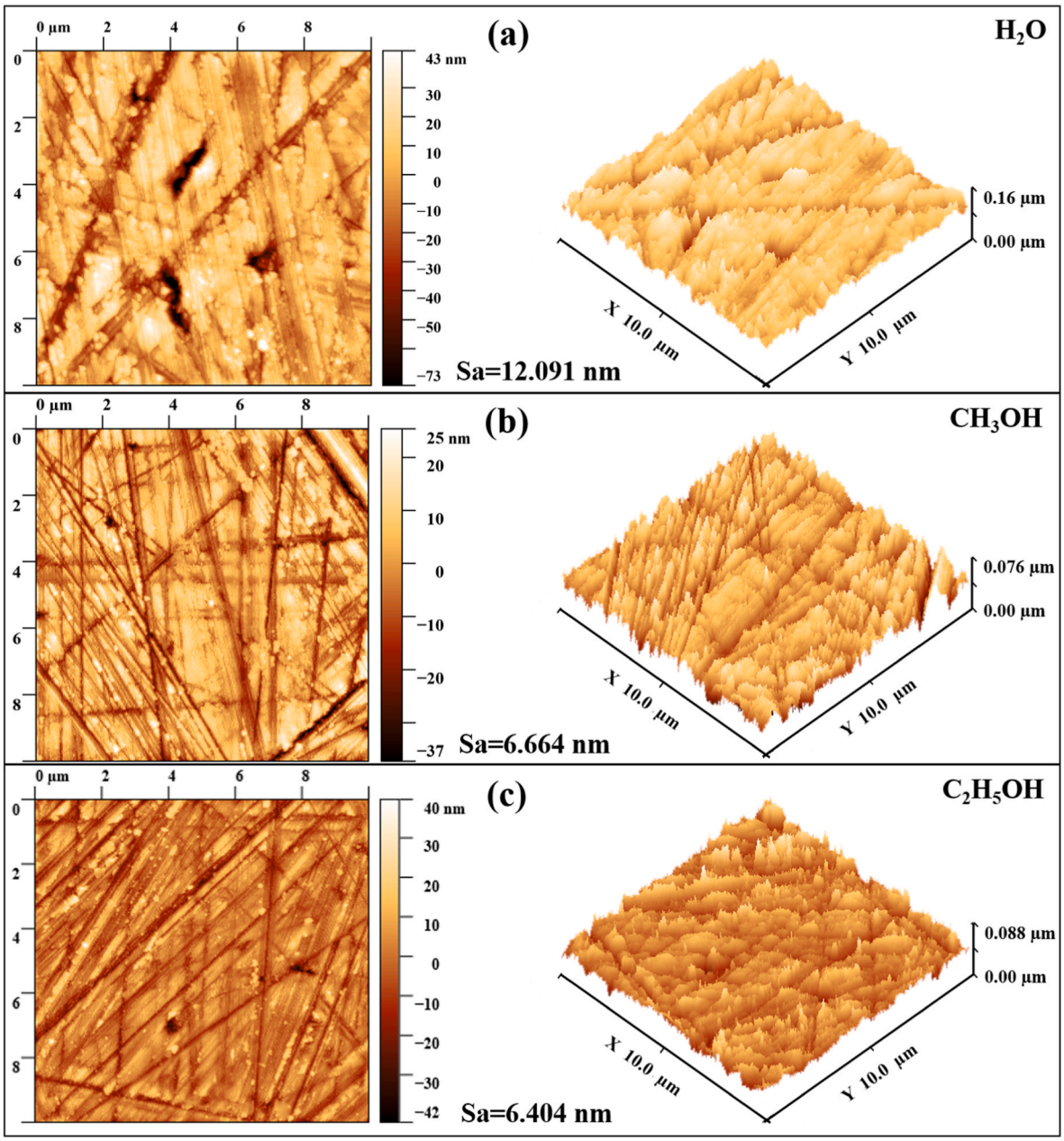

Fig. 12. AFM and 3D topography diagrams after polishing the C face of SiC wafer with different solvents (a) H_2O, (b) CH_3OH, (c) C_2H_5OH.

wafers likely due to the development of a *C*–C layered structure on the C face of SiC. The lubricious effects of alcoholic solvents facilitate the wetting behavior of these graphene-like *C*–C layered structures resulting in lower friction coefficients and wear rates [53], thus promoting the attainment of a smoother surface. Consequently, the C face exhibits a lower Sa. The different surface quality of *C*-side and Si-side of wafers treated with the same solvent further indicates that solvents produce different chemical contributions in polishing *C*-side and Si-side.

3.4.2. *XPS analysis of solvent corroded SiC wafer*

To further determine the chemical reactions between different solvents (H_2O, CH_3OH, and C_2H_5OH) and the SiC wafers, and to verify the molecular dynamics simulation results between the solvents and the SiC wafers. The XPS analysis is performed on the SiC wafer surface corroded by the different solvents to identify their chemical products on the wafer surfaces, as shown in Fig. 14, respectively. The binding energy peaks of C_{1s}, O_{1s}, Si_{2s}, and Si_{2p} photoelectrons on the *C*-face and the Si-face of the SiC wafers corroded by H_2O, CH_3OH, and C_2H_5OH. The ratio of O element to Si element scanned from the *C*-face of the SiC wafer etched by C_2H_5OH is the highest element, and the Si face etched by CH_3OH possesses the similar peak. The elemental ratios of C and Si on the surface of etched SiC wafers by the solvents were higher than the normal stoichiometric ratio of 1, which may be caused by the chemical reactions between the solvents and the SiC surfaces. The relative intensities of the O_{1s} binding energies of the *C*-face and the Si-face of the SiC wafers corroded by the solvents indicate the different chemical reactivity between the solvents and the SiC surfaces.

The fitting analysis of Si_{2p}, C_{1s}, and high-resolution XPS spectrum of the solvent-etched SiC wafer surface is carried out, as shown in Figs. 15 and 16. It can be seen from Fig. 15 that Si_{2p} can be divided into four main peaks: *Si*–C (100.1 eV), *Si*–*C*–O (100.6 eV), SiO_xC_y (101.5 eV), *Si*–O_2 (102.6 eV). It can be seen from Fig. 16 that C1s can be divided into 7 main peaks: *Si*–C (282.4 eV), $Si_4C_{4-x}O_2$ (283.2 eV), *C*–C/C–H (284.6 eV), $Si_4C_4O_4$(285.1 eV), *C*–O (286.3 eV), C═O (288.0 eV), COOH (288.9 eV).

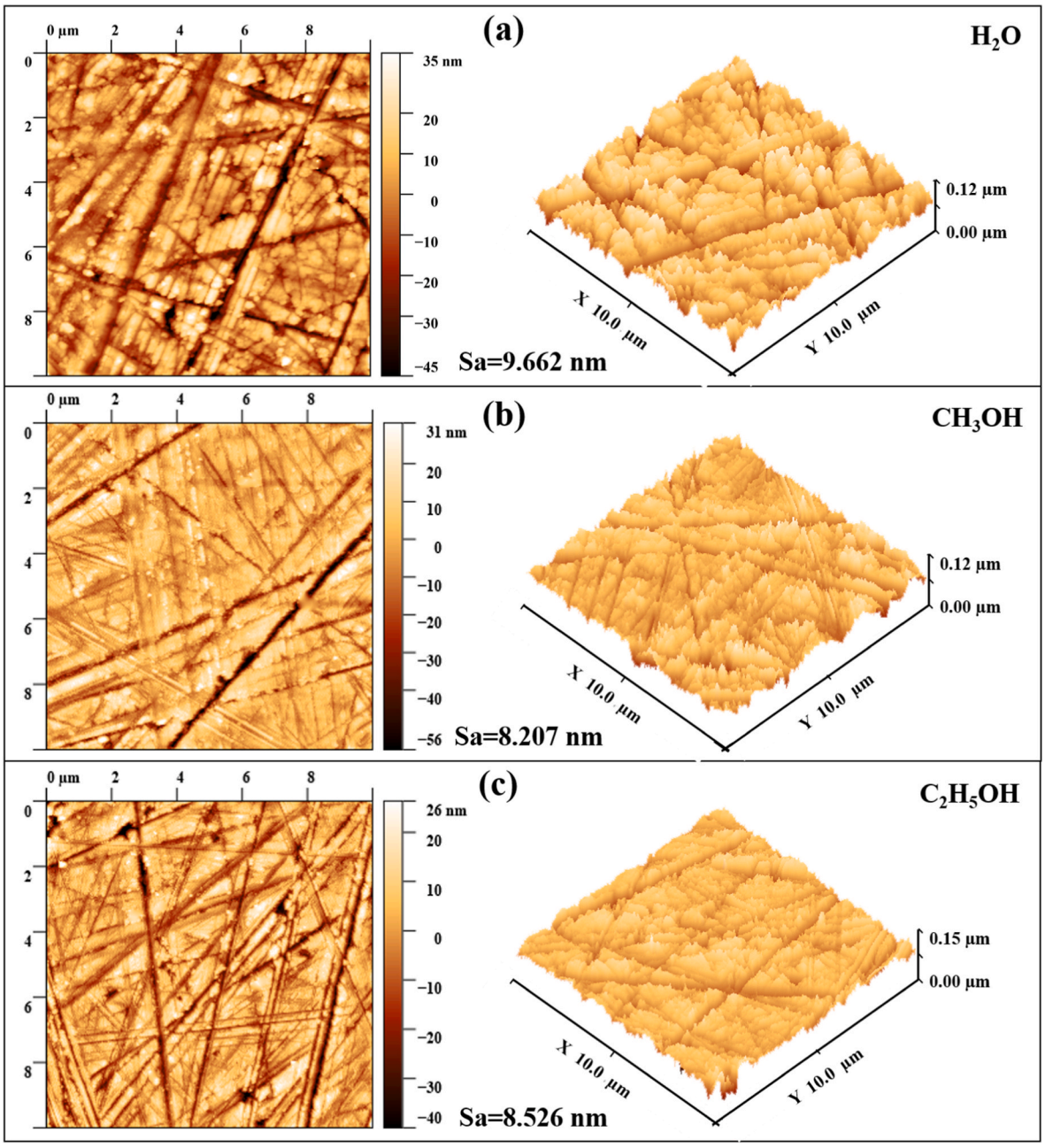

Fig. 13. AFM and 3D topography diagram after polishing the Si face of SiC wafer with different solvents (a) H_2O, (b)CH_3OH, (c) C_2H_5OH.

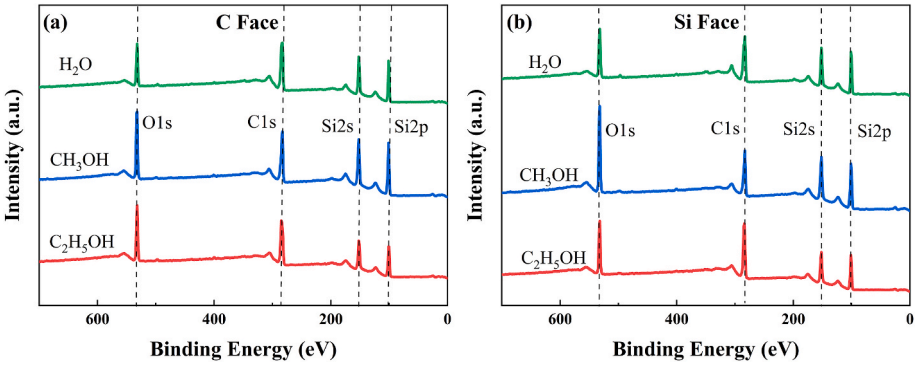

Fig. 14. XPS spectrums of (a) *C*-Face and (b)Si-Face of the SiC wafers etched with different solvents.

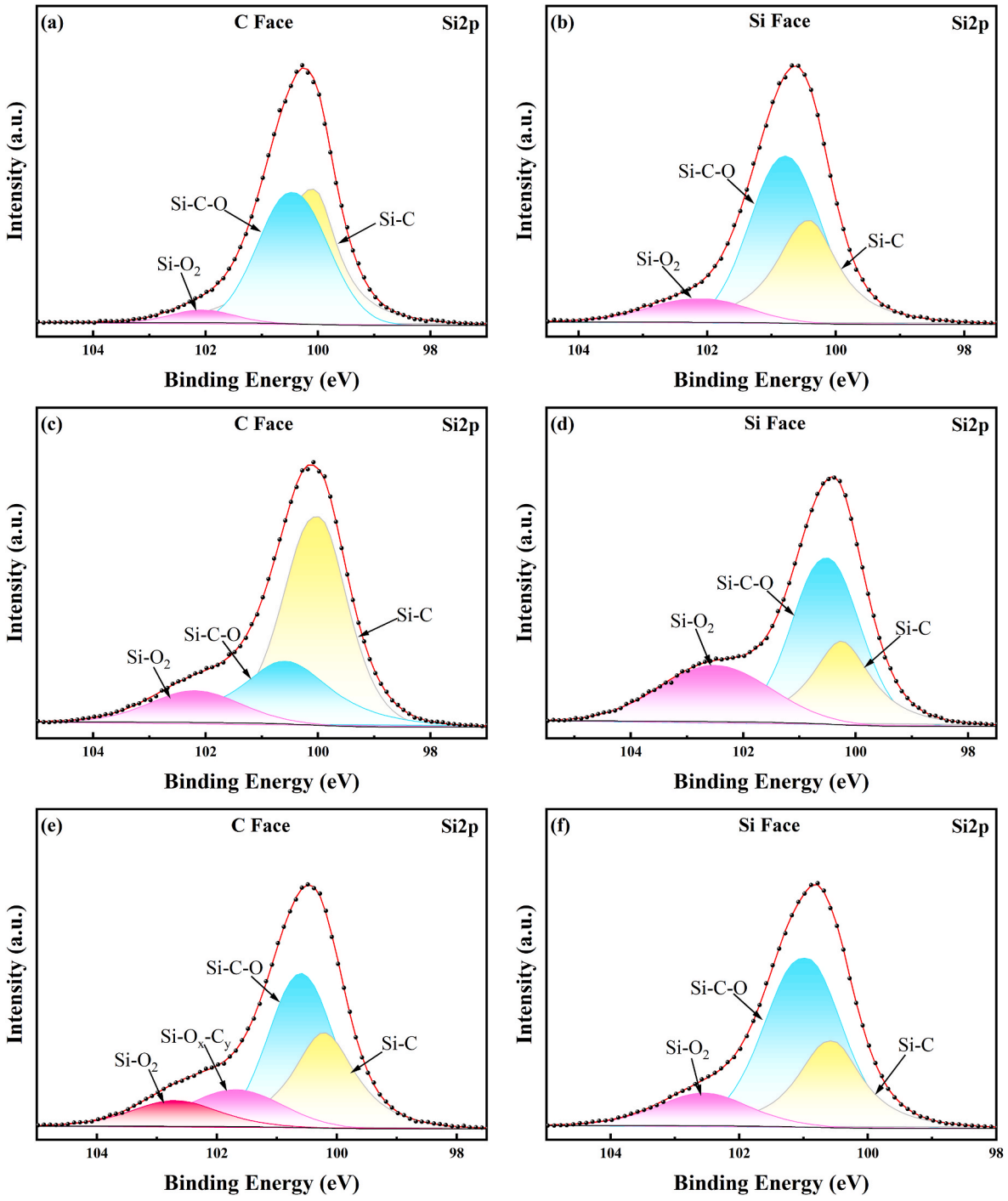

Fig. 15. Si_{2p} spectrums of solvent corroded C face and Si face of SiC substrates consisting of (a) H_2O–C Face, (b) H_2O–Si Face, (c) CH_3OH–C Face, (d) CH_3OH–Si Face, (e) C_2H_5OH–C Face, (f) C_2H_5OH–Si Face.

On the interfacial plane where the solvent interacts with silicon carbide, the relative potency of *Si*–C species in relation to *Si*–*C*–O species serves as a paramount index of the extent of chemical reaction taking place on the corroded surface: an increased strength of *Si*–C species corresponds to a diminished occurrence of chemical reaction. Based on the data presented in Fig. 15, CH_3OH exhibits the least chemical reactivity while C_2H_5OH exhibits the highest chemical reactivity in the context of solvent-substrate interaction with the *C*-face of SiC substrate. This observation aligns well with the observed removal rate of the *C*-face during polishing. Conversely, in the case of solvent interaction with the Si-face of the SiC substrate, H_2O demonstrates the lowest chemical reactivity while C_2H_5OH exhibits the highest chemical reactivity. This result is also in agreement with the observed material removal rate of the Si-face during the three solvent polishing experiments. In summary, the XPS analysis results of three solvents corroding the *C*-face and the Si-face are consistent with the material removal rates of the SiC substrates, which indicates that the chemical reaction between the SiC surface and the solvent leads to the differences in material removal of the SiC substrates polished by the different solvents.

The chemical state of C atoms was confirmed by analyzing the SiC surface corroded by three solvents. Four peaks around 282.4 eV, 283.2 eV, 284.6 eV, and 285.1 eV in Fig. 16 were fitted as the C_{1s} peaks of the

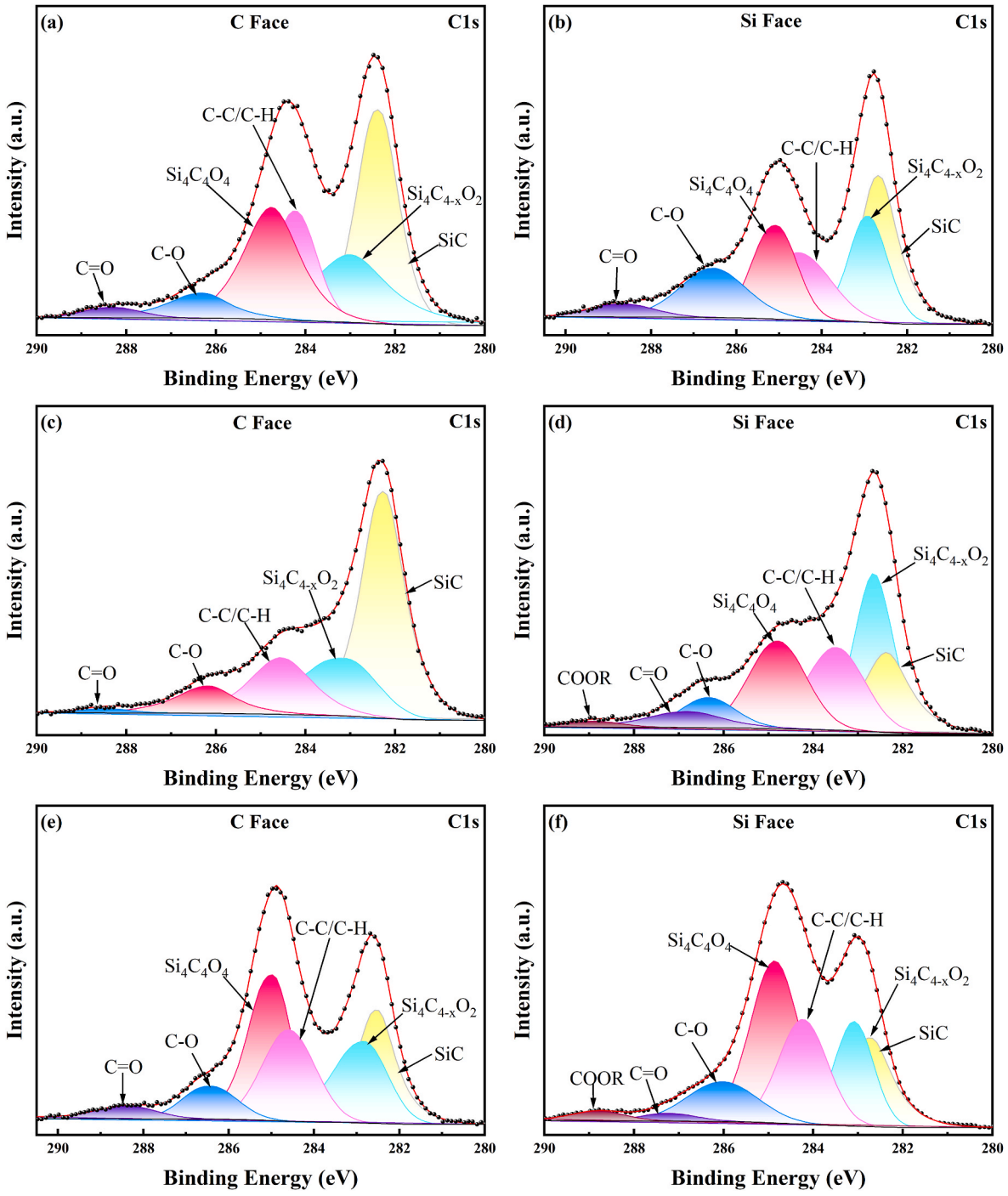

Fig. 16. C_{1s} spectrums of solvent-corroded *C*-face and Si-face of SiC substrates consisting of (a) H_2O–C Face, (b) H_2O–Si Face, (c) CH_3OH–C Face, (d) CH_3OH–Si Face, (e) C_2H_5OH–C Face, (f) C_2H_5OH–Si Face.

SiC substrates etched with different solvents. They correspond to the binding energies to *Si*–C, $Si_4C_{4-x}O_2$, *C*–C/C–H, and $Si_4C_4O_4$. The peaks are close 283.2 eV, 284.6 eV, and 285.1 eV due to the formation of SiC oxides on the SiC surface, which is the oxidation of the SiC surface via solvent corrosion. Careful observation shows that the peak intensity of this part of silicon carbide at 283.2–285.1 eV is sequentially enhanced from the C_{1s} spectrum where SiC is corroded by H_2O to CH_3OH to C_2H_5OH. The interaction strength between the solvent on the Si-face and the *C*-face is C_2H_5OH > CH_3OH > H_2O, which is consistent with the results in the calculation simulation. *C*–O bonds were also observed around 286.3 eV when three solvents were used to polish SiC, suggesting that the chemical attack on the SiC surface comes from the polar –OH groups in the solvents. This is consistent with the results of the transfer of –OH on the SiC surface in the simulations of the previous two sections. These results demonstrate the accuracy of the solvent-SiC surface interaction mechanism in the computational simulation.

Table 3 lists the relative percentages of surface species and chemical components on the *C*-face and Si-face of the SiC substrates, respectively, after interacting with the solvent. In the Si_{2p} spectrum observed on the C face, it is noted that the relative atomic abundance of *Si*–C on the surface of SiC subjected to ethanol corrosion is the least, whereas the relative atomic concentration of *Si*–C on the SiC surface corroded by methanol demonstrates the highest values. The *Si*–C atomic content in the C_{1s} spectrum of the SiC substrate's C face is comparatively minimal in

Table 3
Relative percentage content of chemical composition of the *C*-face of solvent-corroded SiC wafer (%).

Element	Chemical state	H_2O(C)	CH_3OH(C)	C_2H_5OH(C)	H_2O(Si)	CH_3OH(Si)	C_2H_5OH(Si)
Si_{2p}	*Si*–C	49.09	59.85	30.32	35.52	25.58	28.68
	Si–*C*–O	46.69	27.64	44.17	53.73	48.20	57.71
	SiO_xC_y	–	–	15.24	–	–	–
	Si–O_2	4.22	12.51	10.27	10.57	26.22	13.61
C_{1s}	*Si*–C	32.02	31.16	20.25	27.50	17.83	17.79
	$Si_4C_{4-x}O_2$	20.19	34.71	20.55	18.71	26.29	15.79
	C–C/C–H	16.07	15.19	19.87	18.27	20.98	19.84
	$Si_4C_4O_4$	23.76	12.45	27.06	17.70	21.03	31.05
	C–O	5.89	5.81	7.36	14.27	7.06	10.91
	C═O	2.07	0.68	4.92	3.54	5.46	1.68
	COOR	–	–	–	–	1.35	2.95

ethanol. The diminished concentration of *Si*–C substantiates ethanol's superior chemical reactivity. This observation aligns with the enhanced MRR achieved by polishing the SiC substrate's C face using ethanol, as well as the outcomes from simulation calculations. Additionally, analyses of the Si_{2p} and C1s spectra of the SiC surface etched with an alcohol solvent unveiled a higher presence of oxidized species (*Si*–O_2, $Si_4C_{4-x}O_2$, $Si_4C_4O_4$), thereby further confirming the alcohol's capacity for extensive oxidation of the SiC surface as compared to water. The relative intensity of $Si_4C_4O_4$ in the C_{1s} spectrum of the *C*-face and Si-face of solvent-etched SiC corresponds to the MRR of SiC in polishing, reflecting the chemical contribution of solvation to the CMP processes.

4. Conclusions

In this paper, the reaction process of H_2O, CH_3OH, C_2H_5OH solvent molecules on SiC surface is studied by ReaxFF molecular dynamics simulation and polishing experiments. The simulation results show that the solvent molecule undergoes molecular adsorption and dissociative adsorption on the SiC surface, resulting in the surface covered with HO, OH, H, alkyl, alkoxy, and vinyl groups that bond to surface Si or C. The groups adsorbed on the surface of SiC can lead to the *Si*–C bond breaking. As the temperature rises, the surface reaction speeds up. The oxidation of SiC surface in water and alcohol solvent is divided into three stages: the first stage, the solvent molecules dissociate on the SiC surface and occupy the uncoordinated Si atoms and C atoms on the surface; the second stage, the solvent molecules dissociate the groups adsorbed on the surface of SiC, reduce the surface energy and promote the breaking of *Si*–C bonds; in the third stage, the migration of OH and H on the SiC surface induces the formation of *Si*–*O*–Si bonds, which promotes the continuous oxidation of the SiC surface. The results of solvent polishing experiment, atomic force microscope observation and solvent corrosion XPS experiment on SiC surface all support that the chemical activity of alcohols and water on the SiC surface differs significantly, especially on the Si-face of SiC, $C_2H_5OH > CH_3OH > H_2O$, where alcohols demonstrate higher chemical reactivity towards to SiC, validating results derived from ReaxFF simulation. Through these studies, we reveal in detail the mechanism of non-aqueous solvent enhanced material removal rate, and provide options for the use of novel non-aqueous polishing slurries in the third-generation semiconductor of silicon carbide polishing process.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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